

You've been witness to way too many of such incidents and with the dawn of the age of PC and smart devices, you have even more such incidents happening. Perhaps the most relatable example would be the mobile phone heating up when you run some intensive game on it. Those with PCs will recant similar incidents that came to be when they ran an intensive application or game.



This isn't the first time you've done this

The downside of excessive heat

The reason we say that heat and electricity don't go together is because of the destructive effect of heat on electrical gadgets. Heat is a natural byproduct of an electric current passing through a conductor, as we've mentioned previously. However, it has some destructive effects on electrical and electronic components.

Physical effects of heating include the rubber insulation on wires melting off. Then there's the fact that damping components that are used in circuits tend to crack up under excessive heat generation.



Your worst nightmare!

These physical effects when observed at the microscopic level within integrated circuits paints a similar picture. The physical properties of materials within ICs change when subject to excess heat. Certain components which are meant to only allow current to flow in one direction such as a forward-bias PN junction diode will start allowing the current flow way earlier than it is supposed to i.e. cut-off voltage drops. In reverse bias, the opposite happens.

This isn't a new phenomenon. In fact, ICs now have built-in mechanisms to deal with such harmful effects of heating. One of the ways you can experience this is by running an intensive app on your phone while it is cool